



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25

Class: TY	Subject: Reinforced Cement Concrete Structures	Subject Code: PCC-CE-311
Date: 25-06-2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q. No.	Questions	Marks	CO	BT
Q 1	Discuss the difference between limit state method and working stress method.	6	CO1	BT1
Q 2	Define the following terms: a) Factor load b) Characteristics Strength of Concrete c) Creep Strain d) Service load	4	CO1	BT1
Q 3	Explain under reinforced section, limiting section and over-reinforced section with neat sketch of reinforced concrete beam.	6	CO1	BT1
Q 4	What is the bond stress? Write a formula of bond stress and its specification as per IS 456.	4	CO1	BT1
Q 5	Signly reinforced section of a simply supported conceret beam under bending with 4 bars of 16 mm provided in tension zone with dimension of 300 mm X 600 mm overall having effective cover of 45 mm. Assume M20 grade of concrete and Fe415 grade of steel. Determine moment of resistance using limit state method and working stress method. If the beam have 6m length then how much uniformly distributed load beam can carry.	12	CO2	BT 3 BT 4
Q 6	A rectangular beam of size 350x500mm overall depth is simply supported over a span of 7.5m. This beam is subjected to a total service load of 50 kN/m. Design the beam using the limit state method. Take M30 and Fe500 grades of material for design. Assume a 40 mm effective cover on both sides.	10	CO3 CO4	BT3 BT4

Strain	Fsc (N/mm ²)	Strain	Fsc (N/mm ²)
0.00174	348	0.00277	413
0.00195	370	0.00312	424
0.00226	392	0.00415	435

Q 7	Design a simply supported slab using limit state method with span of 6.5m X 3m which supported over 300 mm thick wall. The intensity of live load on slab is 5 kN/m ² and floor finish thickness is assume to be 50mm. Use M20 and Fe 415 grade of material for design.	12	CO3 CO4	BT3 BT4
Q 8	Design a interior slab panel of size 3 m X 4.5 m. The slab is subjected to live load of 8 kN/m ² and floor finishing of 80 mm thick material is required for water proofing of slab. The materials use for the design are M20 and Fe415 grades. Design slab using IS code and limit state method. The slab is supported over 230 mm thick support.	12	CO3 CO4	BT3 BT4
Q 9	Design a axial loaded square column for a service load of 1650 kN. The size of column is 450x450 mm and has effective length of 3500mm. The grade of material used for the design is M25 and Fe415. Use limit state method for design and shows the reinforcement details.	12	CO3 CO4	BT3 BT4
Q 10	Design a isolated footing for a column of size 450 mm X 450 mm subjected to factored load of 1600 kN. Safe bearing capacity of soil is 110 kN/m ² . Use M25 and Fe415 grade of material. Use limit state method for design. Assume the B/L=1 for footing.	12	CO3 CO4	BT3 BT4
Q 11	Design a rectangular beam using working stress method (WSM) for a span of 8m simply supported subjected to live load of 40 kN/m excluding self weight. If the size of beam is to be restricted to 500 mm X 700 mm overall. Use M25 and Fe500 grade of materials. Consider effective cover is 50mm.	10	CO3 CO4	BT3 BT4

Course Outcome (CO)	Bloom's Taxonomy (BT)
CO1 : Understand the basic concepts of structural design methods of R.C.C. and solve problems using Working Stress and Limit State design methods.	BT1-Remember
CO2 : Apply knowledge of structural properties of materials i.e. steel and concrete for assessing the strength of various components of buildings.	BT2- Understand
CO3 : Analyze real problems for collapse and serviceability.	BT3- Apply
CO4: Design various structural elements for compression, tension, shear, torsion and analyze one way and two way slabs.	BT4- Analyze
	BT5 – Evaluate
	BT6 – Create

Give data for the design

Table 19 Design Shear Strength of Concrete, τ_v , N/mm²
(Clauses 40.2.1, 40.2.2, 40.3, 40.4, 40.5.3, 41.3.2, 41.3.3 and 41.4.3)

$100 \frac{A_s}{bd}$	Concrete Grade					
	M 15	M 20	M 25	M 30	M 35	M 40 and above
(1)	(2)	(3)	(4)	(5)	(6)	(7)
≤ 0.15	0.28	0.28	0.29	0.29	0.29	0.30
0.25	0.35	0.36	0.36	0.37	0.37	0.38
0.50	0.46	0.48	0.49	0.50	0.50	0.51
0.75	0.54	0.56	0.57	0.59	0.59	0.60
1.00	0.60	0.62	0.64	0.66	0.67	0.68
1.25	0.64	0.67	0.70	0.71	0.73	0.74
1.50	0.68	0.72	0.74	0.76	0.78	0.79
1.75	0.71	0.75	0.78	0.80	0.82	0.84
2.00	0.71	0.79	0.82	0.84	0.86	0.88
2.25	0.71	0.81	0.85	0.88	0.90	0.92
2.50	0.71	0.82	0.88	0.91	0.93	0.95
2.75	0.71	0.82	0.90	0.94	0.96	0.98
3.00 and above	0.71	0.82	0.92	0.96	0.99	1.01

NOTE — The term A_s is the area of longitudinal tension reinforcement which continues at least one effective depth beyond the section being considered except at support where the full area of tension reinforcement may be used provided the detailing conforms to 26.2.2 and 26.2.3

Table 20 Maximum Shear Stress, $\tau_{v \max}$, N/mm²
(Clauses 40.2.3, 40.2.3.1, 40.5.1 and 41.3.1)

Concrete Grade	M 15	M 20	M 25	M 30	M 35	M 40 and above
$\tau_{v \max}$, N/mm ²	2.5	2.8	3.1	3.5	3.7	4.0

26.2.1.1 Design bond stress in limit state method for plain bars in tension shall be as below:

Grade of concrete	M 20	M 25	M 30	M 35	M 40 and above
Design bond stress, τ_{bd} , N/mm ²	1.2	1.4	1.5	1.7	1.9

**Table 26 Bending Moment Coefficients for Rectangular Panels Supported on
Four Sides with Provision for Torsion at Corners**
(Clauses D-1.1 and 24.4.1)

Case No.	Type of Panel and Moments Considered	Short Span Coefficients α_x (Values of l_y/l_x)								Long Span Coefficients α_y for All Values of
		1.0	1.1	1.2	1.3	1.4	1.5	1.75	2.0	l_y/l_x
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
1	<i>Interior Panels:</i>									
	Negative moment at continuous edge	0.032	0.037	0.043	0.047	0.051	0.053	0.060	0.065	0.032
	Positive moment at mid-span	0.024	0.028	0.032	0.036	0.039	0.041	0.045	0.049	0.024
2	<i>One Short Edge Continuous:</i>									
	Negative moment at continuous edge	0.037	0.043	0.048	0.051	0.055	0.057	0.064	0.068	0.037
	Positive moment at mid-span	0.028	0.032	0.036	0.039	0.041	0.044	0.048	0.052	0.028
3	<i>One Long Edge Discontinuous:</i>									
	Negative moment at continuous edge	0.037	0.044	0.052	0.057	0.063	0.067	0.077	0.085	0.037
	Positive moment at mid-span	0.028	0.033	0.039	0.044	0.047	0.051	0.059	0.065	0.028
4	<i>Two Adjacent Edges Discontinuous:</i>									
	Negative moment at continuous edge	0.047	0.053	0.060	0.065	0.071	0.075	0.084	0.091	0.047
	Positive moment at mid-span	0.035	0.040	0.045	0.049	0.053	0.056	0.063	0.069	0.035
5	<i>Two Short Edges Discontinuous:</i>									
	Negative moment at continuous edge	0.045	0.049	0.052	0.056	0.059	0.060	0.065	0.069	—
	Positive moment at mid-span	0.035	0.037	0.040	0.043	0.044	0.045	0.049	0.052	0.035
6	<i>Two Long Edges Discontinuous:</i>									
	Negative moment at continuous edge	—	—	—	—	—	—	—	—	0.045
	Positive moment at mid-span	0.035	0.043	0.051	0.057	0.063	0.068	0.080	0.088	0.035
7	<i>Three Edges Discontinuous (One Long Edge Continuous):</i>									
	Negative moment at continuous edge	0.057	0.064	0.071	0.076	0.080	0.084	0.091	0.097	—
	Positive moment at mid-span	0.043	0.048	0.053	0.057	0.060	0.064	0.069	0.073	0.043
8	<i>Three Edges Discontinuous (One Short Edge Continuous):</i>									
	Negative moment at continuous edge	—	—	—	—	—	—	—	—	0.057
	Positive moment at mid-span	0.043	0.051	0.059	0.065	0.071	0.076	0.087	0.096	0.043
9	<i>Four Edges Discontinuous:</i>									
	Positive moment at mid-span	0.056	0.064	0.072	0.079	0.085	0.089	0.100	0.107	0.056



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25 (Semester – II)

Class: Third Year	Subject: Geotechnical Engineering	Subject Code: PCC-CE310
Date: 18.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- i. Mobile phones and programmable calculators are strictly prohibited.
- ii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iii. Figures to the right indicate full marks.

Q.N.	Questions	Marks	CO	BT
Q.1	Explain the following with neat sketch : 1) Finite and infinite slope. a) 2) Inside clearance, outside clearance and area ratio of sampler. 3) Wash boring.	15	3,4	3
	Explain the following with neat sketch : b) 1) Liquid limit test. 2) Heavy compaction test with large size mould.	10	1,5	2,3
Q.2	When an unconfined compression test is conducted on a cylinder of soil, it fails under an axial stress of 1.5 kg/cm^2 . The failure plane makes an angle of 60° with the horizontal. Determine the cohesion and angle of internal friction.	05	2	4
Q.3	A clay soil, tested in a consolidometer, showed a decrease in void ratio from 1.20 to 1.10 when the pressure was increased from 24.51 to 49.03 kN/m^2 . Calculate the coefficient of compressibility and the coefficient of volume compressibility.	05	2	4
Q.4	Draw the standard plasticity chart to classify the fine-grained soil according to unified soil classification (USCS) and Indian standard classification system (ISCS).	05	4	2
Q.5	Explain falling head permeability test with neat sketch and derive the formula for the same.	10	2	2,3
Q.6	Derive an expression for effective stresses at top and bottom of the soil surface with neat sketch for the flow of water in no flow condition and downward condition.	10	2	3

Q.7	The following results were obtained from the sand replacement method. Mass of soil excavated from the hole = 6 kg Water content of soil = 22 % Mass of dry soil to fill a hole = 4.2 kg Mass of dry sand to fill the container = 6.2 kg Volume of the container = 4.8 litres Calculate the wet and dry densities of the soil, if the specific gravity of the particles is 2.7.						10	5	4														
Q.8	The following are the results of a compaction test <table border="1"><tr><td>Weight of wet soil (N)</td><td>16.7</td><td>18.54</td><td>19.92</td><td>19.52</td><td>19.23</td><td>18.83</td></tr><tr><td>Water content (%)</td><td>7.7</td><td>11.5</td><td>14.6</td><td>17.5</td><td>19.5</td><td>21.2</td></tr></table> If the volume of compaction mould is 950 cc and specific gravity of solids is 2.65. Find the compaction curve showing the optimum moisture content and maximum dry density.						Weight of wet soil (N)	16.7	18.54	19.92	19.52	19.23	18.83	Water content (%)	7.7	11.5	14.6	17.5	19.5	21.2	10	2	4
Weight of wet soil (N)	16.7	18.54	19.92	19.52	19.23	18.83																	
Water content (%)	7.7	11.5	14.6	17.5	19.5	21.2																	
Q.9	A soil sample has a porosity of 40 %. The specific gravity of solids is 2.70. Calculate (a) void ratio (b) dry density (c) unit weight if the soil is 50 % saturated and (d) unit weight if the soil is completely saturated.						10	1	4														
Q.10	A retaining wall, 7.5 m high, retains a cohesionless backfill. The top 3 m of the fill has a unit weight of 18 kN/m ³ and $\phi = 30^\circ$ and the rest has a unit weight of 24 kN/m ³ and $\phi = 20^\circ$. Determine the lateral active earth forces per meter run of the wall at A, B and C and its point of application with diagrams.						10	3	4														

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand and determine Index properties of soil.
CO 2	Understand the permeability, compaction, shear strength and consolidation of soil and its evaluation.
CO 3	Analyse problems related to slope failures and calculate earth pressure for different soil under various conditions.
CO 4	Classify the soil and comprehend different methods of soil exploration for preparation of soil profile.
CO 5	Gain mastery in experimental determination of different properties of soil.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
Summer term Examination
Academic Year: 2024-25

Class: TY	Subject: Engineering Geology	Subject Code: AUD-CE308
Date: 24.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- All questions are compulsory

Q.N.		Marks	CO	BT
Q.1	a Explain i) Types of dams (with diagram) ii) Geological conditions for site selection of dam.	10	CO5	BT3
Q.2	a What is landslide? Explain types of landslides, causes of landslides and their effects.	10	CO5	BT2
Q.3	a What is earthquake? Explain types of seismic waves, causes of earthquake and earthquake prone zones.	10	CO5	BT2
Q.4	a Explain the following i) What is fold ii) Terminologies of fold iii) Types of fold (with diagram)	10	CO2	BT2
Q.5	a What is groundwater? Explain i) Types of groundwater ii) Types of aquifers	10	CO4	BT2
Q.6	a Explain the following iv) What is fault v) Terminologies of fault Types of faults (with diagram)	10	CO3	BT2
Q.7	a Classify tunnel based on shape (with application and diagram) and explain geological consideration at tunnel site.	10	CO5	BT1
Q.8	a What is weathering? explain types of weathering.	10	CO1	BT1
Q.9	a Explain types of landforms.	10	CO3	BT1
Q.10	a Write a note on metamorphic rock.	10	CO1	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	On successful completion of this course, students should be able to:
CO 2	Understand weathering process and mass movement
CO 2	Distinguish geological formations



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25

Class: T.Y. B. Tech.

Course: Engg. Hydrology

Course Code: PCC-CE303

Date: 17.06.2025

Time: 10 a. m. to 01:00 p.m.

Max. Marks: 100

Note:

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- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Name of Question	Marks	CO	BT
Q.1	a What are the different forms of precipitation? Describe in detail about Symons raingauge.	10	CO1	BT1
	b Discuss the double mass curve technique used for testing consistency of rainfall data. Explain the Hyetograph used for presentation of rainfall data.	10	CO1	BT2
Q.2	a Explain about following infiltration indices: (1) ϕ index (2) W-index.	10	CO2	BT2
	b Explain the Baney-Cridle formula used to estimate potential evapotranspiration. State the factors affecting infiltration capacity of soil.	10	CO2	BT3
Q.3	a Explain the rational method used for estimation of runoff. Discuss about flow duration curve.	10	CO3	BT2
	b How the runoff volume is expressed using water balance equation? Explain the flow -mass curve.	10	CO3	BT3
Q.4	a Define the term Effective Rainfall. Explain the different methods of base flow separation.	10	CO4	BT2
	b State and explain the assumptions made in unit hydrograph theory. The 6-h unit hydrograph of a catchment of area 1000 km ² can be approximated as a triangle with the base of 60 h. Calculate the peak ordinate of this unit hydrograph.	10	CO4	BT4
Q.5	a Define following terms: (1) Time of concentration (2) Runoff coefficient. Explain the rational method used to calculate peak runoff volume.	10	CO5	BT2
	b Explain the Gumbel's method. For a river the estimated flood peaks for two return periods by Gumbel's method are as follows: Return Period (Years): 100 50 Peak Flood (m ³ /s) 435 395 Calculate flood discharge for 200 years return period.	10	CO5	BT4

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:			
CO 1	Analyze precipitation data.	CO 2	Understand infiltration and evapotranspiration process.
CO 3	Determine runoff volume.	CO 4	Comprehend unit hydrograph theory.
CO 5	Understand flood frequency analysis.		

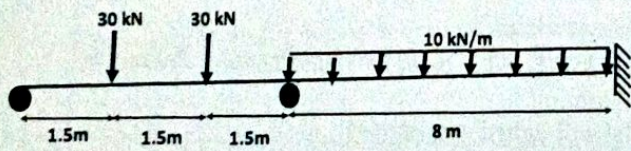
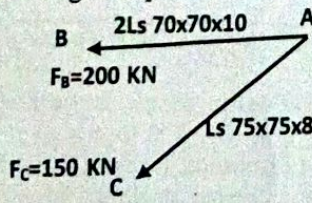


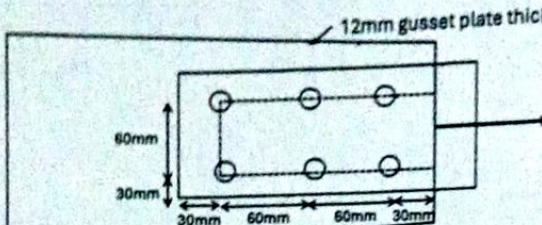
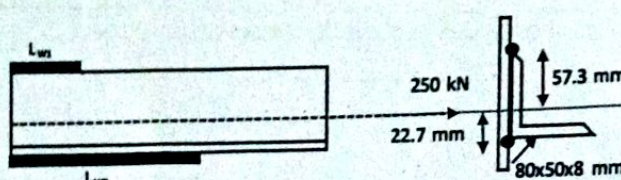
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25

Class: TY	Subject: Design of Steel Structures	Subject Code: PCC-CE-301
Date: 16-06-2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Question statement	Marks	CO	BT
Q.1	Discuss the types of steel and also explain the stress-strain curve for the mild steel.	6	CO1	BT1 BT2
Q.2	Define the following terms a) Plastic moment b) Elastic moment c) Shape factor	6	CO1	BT1 BT2
Q.3	Determine the shape factor for the following shapes: a) Rectangular section of size BxD. b) Solid circular section of diameter D.	8	CO1	BT2 BT3
Q.4	A beam is subjected to a collapse load system, as shown in Figure. If the beam is of uniform cross-section throughout its length, determine the plastic moment capacity required for the beam. 	10	CO2	BT4
Q.5	Design the connection for joint A of the roof truss as shown in Figure. The members are connected with 16 mm diameter bolts of grade 4.6 to the gusset plate 12 mm thick. 	10	CO3 CO4	BT5
Q.6	Discuss the steps to design laterally supported beams in details.	10	CO3 CO4	BT1

Q. 7	Determine the design tensile strength of the plate 120 mm x 8 mm connected to a 12 mm thick gusset plate with bolt holes as shown in Figure. The yield strength and ultimate strength of the steel used are 250 MPa and 400 MPa. The diameter of the bolts used is 16 mm. 	10	CO2	BT1, BT2
Q. 8	A tie member consisting of an ISA 80mm X 50mm X 8mm is welded to a 12mm thick gusset plate at the site. Design a weld on two sides of angle, as shown in Figure. to transmit a factored load of 250 kN. Consider Fe410 grade of steel for design. 	10	CO3	BT3 BT4
Q. 9	Design a builtup section for a column of 10 m long carries a factored axial load of 1000kN with single lacing system having bolted connections. Assume steel of grade Fe410 and bolts of grade 4.6. The builtup column is design using two I section placed back to back. Shows all the details in neat sketch. Assume design compressive stress (f_{cd}) is 150 N/mm².	15	CO3 & CO4	BT3, BT4 & BT5
Q.10	Design a suitable bolted gusseted base for a column ISHB 350@ 661.2 N/m carries an axial compressive load of 1750 kN. The base rests on M15 grade concrete pedestal. Use 24 mm diameter bolts of grade 4.6 for the connections. Properties of ISHB 350@ 661.2 N/m; $D=350$ mm; $b_f = 250$mm; $t_f = 11.6$ mm; $t_w=8.3$mm. Shows all the details of bolted connections.	15	CO3 & CO4	BT3, BT4 & BT5

== ALL THE BEST ==

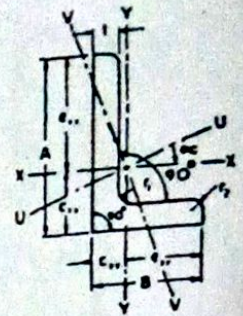
Course Outcome (CO)	Bloom's Taxonomy (BT)
CO1 : Practice fundamentals of strength of materials for design of steel structural elements.	BT1-Remember
CO2 : Understand fundamentals of load assessment and design criteria for various structural steel elements.	BT2- Understand
CO3 : Apply design philosophy of limit state method to structural fasteners, tension members, compression members, simple and built-up beams, single and built-up columns as per given loading conditions.	BT3- Apply
CO4: Analyzed and design structural fasteners, tension members, compression members, simple and built-up beams, single and built-up columns for real problem.	BT4- Analyze
CO5 : Evaluate the performance of any steel structure subjected to vertical loading.	BT5 – Evaluate
	BT6 – Create

Table Contd.)

Designation	Weight per Meter N/m	Sectional Area (A) cm ²	Depth of Section (h) mm	Width of Flange (b _f) mm	Thickness of Flange (t _f) mm	Thickness of Web (t _w) mm	R ₁ mm	R ₂ mm	Radius of Gyration (r _y)	(r _x)	I _x cm ⁴	I _y cm ⁴	Section Modulus (Z _x) cm ³	Plastic Modulus (Z _p) cm ³	Shape Factor (Z _p /Z _x)	Depth of Web (d = h - 2 (t _f + R ₁))
ISHB 250	536.61	69.71	250	250	9.7	8.8	10.0	5.0	10.70	5.37	8983.9	2011.7	638.7	708.43	1.1092	210.6
ISLB 325	*422.81	54.09	325	165	9.8	7.0	16.0	8.0	13.41	3.05	9874.6	510.8	607.7	687.76	1.1317	273.4
ISHB 250	500.31	64.96	250	250	9.7	6.9	10.0	5.0	10.91	5.49	7983.9	2011.7	618.9	678.73	1.0967	210.6
ISMC 350	*413.00	53.66	350	100	13.5	8.1	14.0	7.0	13.66	2.83	10008.0	430.6	571.9	672.19	1.1754	295
ISMB 300	*433.60	56.26	300	140	12.4	7.5	14.0	7.0	12.37	2.84	8603.6	453.9	573.6	651.74	1.1362	247.2
ISLC 350	*380.63	49.47	350	100	12.5	7.4	13.0	6.0	13.72	2.82	9312.6	394.6	532.1	622.95	1.1707	299
ISLB 300	*369.84	48.08	300	150	9.4	6.7	15.0	7.5	12.35	2.80	7332.9	376.2	488.9	554.32	1.1338	251.2
ISHB 225	459.11	59.66	225	225	9.1	8.6	10.0	5.0	9.58	4.84	5478.8	1396.6	475.4	527.57	1.1097	212
ISWB 250	401.23	52.05	250	200	9.0	6.7	10.0	5.0	10.69	4.06	5943.1	857.5	469.3	515.82	1.0987	186.8
ISHB 225	422.81	54.94	225	90	13.6	7.6	13.0	6.5	11.81	2.61	5478.8	1396.6	424.2	496.77	1.1711	246.8
ISMC 300	*351.20	45.64	300	125	12.5	6.9	13.0	6.5	10.39	2.65	6362.6	310.8	410.5	465.71	1.1345	199
ISMB 250	365.91	47.55	250	125	12.5	6.9	13.0	6.5	10.39	2.65	5131.6	334.5	403.2	466.73	1.1576	252.8
ISLC 300	*324.71	42.11	300	100	11.6	6.7	12.0	6.0	11.98	2.87	6047.9	346.0	392.4	443.09	1.1305	229.4
ISLB 275	*323.73	42.02	275	140	8.8	6.4	14.0	7.0	11.31	2.61	5375.3	287.0	372.2	414.23	1.1129	164
ISHB 200	392.4	50.94	200	200	9.0	7.8	9.0	4.5	8.55	4.42	3721.8	994.6	360.8	397.23	1.1010	164
ISHB 200	365.91	47.54	200	200	9.0	6.1	9.0	4.5	8.71	4.51	3608.4	967.1	348.5	389.93	1.1189	187.2
ISWB 225	332.56	43.24	225	150	9.9	6.4	9.0	4.5	9.52	3.22	3920.5	448.6	305.3	356.72	1.1684	197.8
ISMC 250	*298.22	38.67	250	80	14.1	7.1	12.0	6.0	9.94	2.38	3816.8	219.8	305.9	348.27	1.1385	177.4
ISMB 225	306.07	39.72	225	110	11.8	6.5	12.0	6.0	9.31	2.34	3441.8	218.3	297.4	338.69	1.1388	207.6
ISLB 250	*273.70	35.53	250	100	10.7	6.1	11.0	5.5	10.23	2.33	3717.8	193.4	295.0	338.11	1.1462	206.6
ISLC 250	274.68	35.65	250	100	10.7	6.1	9.0	4.5	10.17	2.89	3687.9	298.4	262.5	293.99	1.1200	164
ISWB 200	282.53	36.71	200	140	9.0	6.1	12.0	6.0	9.03	2.38	2624.5	328.8	239.5	277.93	1.1605	176.2
ISMC 225	*254.08	33.01	225	80	12.4	6.4	11.0	5.5	9.14	2.62	2547.9	209.5	226.5	260.13	1.1485	182.6
ISLC 225	*235.44	30.53	225	90	10.2	5.8	12.0	6.0	9.15	1.94	2501.9	112.7	222.4	254.72	1.1453	183.8
ISLB 225	*230.54	29.92	225	100	8.6	5.8	11.0	5.5	8.32	2.15	2235.4	150.0	223.5	253.86	1.1358	156.4
ISMB 200	249.17	32.33	200	100	10.8	5.7	11.0	4.0	6.09	3.35	1635.6	494.9	218.1	251.64	1.1538	116
ISHB 150	339.43	44.08	150	150	9.0	8.4	8.0	4.0	6.29	3.44	1540.0	460.3	205.3	232.52	1.1326	116
ISHB 150	300.19	38.98	150	150	9.0	5.4	8.0	4.0	6.50	3.54	1455.6	431.7	194.1	215.64	1.1110	116
ISHB 150	265.85	34.48	150	75	11.4	6.1	11.0	5.5	8.03	2.23	1819.3	140.4	181.9	211.25	1.1614	155.2
ISMC 200	*216.80	28.21	200	75	11.4	6.1	11.0	5.5	8.03	2.23	1819.3	140.4	181.9	211.25	1.1614	155.2

(Table Contd.)

TABLE IV ROLLED STEEL UNEQUAL ANGLES
DIMENSIONS AND PROPERTIES



Designation	Size A x B	Thick- ness t	Sectional Area a	Weight per Metre w	Centre of Gravity		Distance of Extreme Fibre		Moments of Inertia			
					C_{xx}	C_{yy}	e_{xx}	e_{yy}	I_{xx}	I_{yy}	I_{xy}	I_{yy}
	mm mm	mm	cm ²	kg	cm	cm	cm	cm	cm ⁴	cm ⁴	cm ⁴	cm ⁴
ISA 3020	30 x 20	3.0	1.41	1.1	0.98	0.49	2.02	1.51	1.2	0.4	1.4	0.2
		4.0	1.84	1.4	1.02	0.53	1.98	1.47	1.5	0.5	1.8	0.3
		5.0	2.25	1.8	1.06	0.57	1.94	1.43	1.9	0.6	2.1	0.4
ISA 4025	40 x 25	3.0	1.88	1.5	1.30	0.57	2.70	1.93	3.0	0.9	3.3	0.5
		4.0	2.46	1.9	1.35	0.62	2.65	1.88	3.8	1.1	4.3	0.7
		5.0	3.02	2.4	1.39	0.66	2.61	1.84	4.6	1.4	5.1	0.8
		6.0	3.56	2.8	1.43	0.69	2.57	1.81	5.4	1.6	5.9	1.0
ISA 4530	45 x 30	3.0	2.18	1.7	1.42	0.69	3.08	2.31	4.4	1.5	5.0	0.9
		4.0	2.86	2.2	1.47	0.73	3.03	2.27	5.7	2.0	6.5	1.1
		5.0	3.52	2.8	1.51	0.77	2.99	2.23	6.9	2.4	7.9	1.4
		6.0	4.16	3.3	1.55	0.81	2.95	2.19	8.0	2.8	9.2	1.7
ISA 5030	50 x 30	3.0	2.34	1.8	1.63	0.65	3.37	2.35	5.9	1.6	6.5	1.0
		4.0	3.07	2.4	1.68	0.70	3.33	2.30	7.7	2.1	8.5	1.2
		5.0	3.78	3.0	1.72	0.74	3.28	2.26	9.3	2.5	10.3	1.5
		6.0	4.47	3.5	1.76	0.78	3.24	2.22	10.9	2.9	11.9	1.8
ISA 6040	60 x 40	5.0	4.76	3.7	1.95	0.96	4.05	3.04	16.9	6.0	19.5	3.4
		6.0	5.65	4.4	1.99	1.00	4.01	3.00	19.9	7.0	22.8	4.0
		8.0	7.37	5.8	2.07	1.08	3.93	2.92	25.4	8.0	29.0	5.2
ISA 6545	65 x 45	5.0	5.26	4.1	2.07	1.08	4.43	3.42	22.1	8.6	25.9	4.8
		6.0	6.25	4.9	2.11	1.12	4.39	3.38	26.0	10.1	30.4	5.7
		8.0	8.17	6.4	2.19	1.20	4.31	3.30	33.2	12.8	38.7	7.4
ISA 7045	70 x 45	5.0	5.52	4.3	2.27	1.04	4.73	3.46	27.2	8.8	30.9	5.1
		6.0	6.56	5.2	2.32	1.09	4.68	3.41	32.0	10.3	36.3	6.0
		8.0	8.58	6.7	2.40	1.16	4.60	3.34	41.0	13.1	46.3	7.8
		10.0	10.52	8.3	2.48	1.24	4.52	3.26	49.3	15.6	55.4	9.5
ISA 7550	75 x 50	5.0	6.02	4.7	2.39	1.16	5.11	3.84	34.1	12.2	39.4	6.9
		6.0	7.16	5.6	2.44	1.20	5.06	3.80	40.3	14.3	46.4	8.2
		8.0	9.38	7.4	2.52	1.28	4.98	3.72	51.8	18.3	59.4	10.6
		10.0	11.52	9.0	2.60	1.36	4.90	3.64	62.3	21.8	71.2	12.9
ISA 8050	80 x 50	5.0	6.27	4.9	2.60	1.12	5.40	3.88	40.6	12.3	45.7	7.2
		6.0	7.46	5.9	2.64	1.16	5.36	3.84	48.0	14.4	53.9	8.5
		8.0	9.78	7.7	2.73	1.24	5.27	3.76	61.9	18.5	69.3	11.0
		10.0	12.02	9.4	2.81	1.32	5.19	3.68	74.7	22.1	83.3	13.5



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25

Class: SY

Course: Hydraulic Engineering

Course Code: PCC-CE 206

Date: 21.06.2025

Time: 14:30 to 17:30

Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Define following terms: (i) Sub-critical, Critical and Super Critical flow (ii) Non-uniform flow, Rapidly varied flow, Gradually varied flow	05	CO3	BT1
	b	A trapezoidal channel has side slopes of 3 horizontal to 4 vertical and slope of its bed is 1 in 2000. Determine the optimum dimensions of the channel, if it is to carry water at 0.5 m ³ /s. Take Chezy's constant as 80.	06	CO3	BT3
	c	Derive the expression for loss of energy due to hydraulic jump.	06	CO3	BT3
Q.2	a	What do you mean by Prandtl's maxing length theory?	05	CO4	BT2
	b	Explain the hydrodynamically smooth and rough boundaries.	04	CO4	BT2
	c	Describe the step by step procedure of Hardy Cross method for pipe network problems.	08	CO4	BT2
Q.3	a	Define: (i) Boundary Layer Thickness; (ii) Displacement Thickness; (iii) Momentum Thickness; (iv) Energy Thickness	08	CO3	BT2
	b	Find the momentum thickness and energy thickness for the velocity distribution in the boundary layer given by $\frac{u}{U} = \frac{y}{\delta}$, where u is the velocity at a distance y from the plate and u = U at y = δ , where δ = boundary layer thickness. Also calculate the value of $\frac{\theta}{\delta^{**}}$	08	CO3	BT3
Q.4	a	The pressure difference Δp in a pipe of diameter D and length l due to viscous flow depends on the velocity V , viscosity μ and density ρ . Using Buckingham's π -theorem, obtain an expression for Δp .	08	CO4	BT3
	b	What is similarity laws? Explain Reynold's model law.	04	CO4	BT2
	c	Define : (i) Mach number; (ii) Euler's number; (iii) Weber's number; (iv) Froude's number	04	CO1	BT2
Q.5	a	Derive the expression of work done by the centrifugal pump on water	06	CO2	BT3
	b	Define : (i) Manometric Efficiency (ii) Mechanical efficiency (iii) Overall efficiency	03	CO2	BT2
	c	Two geometrically similar pumps are running at the same speed of 1000 r.p.m. One pump has an impeller diameter of 0.30 meter	08	CO2	BT3

		and lifts water at the rate of 20 litres per second against a head of 15 meters. Determine the head and impeller diameter of the other pump to deliver half the discharge.			
Q.6	a	Draw the neat diagram of reciprocating pumps and list the all components of it.	05	CO2	BT2
	b	What is submersible pump? Write the working principle of submersible pump	04	CO2	BT2
	c	A double acting reciprocating pump, running at 40 r.p.m. is discharging 1 m^3 of water per minute. The pump has stroke of 400 mm. The diameter of piston is 200 mm. The delivery and suction heads are 20 m and 5 m respectively. Find the slip of the pump and power required to drive the pump.	08	CO2	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Define the basic concept used in different types of incompressible fluid flow including the working principles of fluid machinery.
CO 2	Understand and apply basics related to turbines & pumps in water resources planning.
CO 3	Apply various components of fluid flow and comparison of possible solutions.
CO 4	Analyzed appropriate concept and apply the same using basic principles for solving incompressible fluid flow problems

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Civil Engineering Summer term Examination Academic Year: 2024-25		
Class: SY	Subject: Surveying and Geomatics	Subject Code: PCC--CE -203	
Date: 25.06.2025	Time: 14:30 to 17:30	Max. Marks: 100	

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- All questions are compulsory

Q.N.		Marks	CO	BT																		
Q.1	a Two tangents AB and BC intersect at a point B at chainage 150.5 m. The curve of radius 100 m and deflection angle 30° is set out by the method of offsets from the long chord. (draw diagram of curve) Calculate i) Tangent length ii) Curve length iii) Length of long chord iv) Mid-ordinate	10	CO4	BT4																		
Q.2	a The following table gives the lengths and bearings of the lines of a closed traverse ABCD. The length and bearing of the line DA have been omitted. Calculate the length and bearing of the line DA. <table><tr><th>Line</th><th>Length (m)</th><th>Bearing</th></tr><tr><td>AB</td><td>100</td><td>$45^\circ 0'$</td></tr><tr><td>BC</td><td>120</td><td>$135^\circ 0'$</td></tr><tr><td>CD</td><td>110</td><td>$225^\circ 0'$</td></tr><tr><td>DA</td><td>?</td><td>?</td></tr></table>	Line	Length (m)	Bearing	AB	100	$45^\circ 0'$	BC	120	$135^\circ 0'$	CD	110	$225^\circ 0'$	DA	?	?	10	CO3	BT4			
Line	Length (m)	Bearing																				
AB	100	$45^\circ 0'$																				
BC	120	$135^\circ 0'$																				
CD	110	$225^\circ 0'$																				
DA	?	?																				
Q.3	a Part A: Convert Whole Circle Bearings (WCB) to Quadrantal Bearings (QB) <table><tr><th>No.</th><th>WCB ($^\circ$)</th><th>QB (Quadrantal Bearing)</th></tr><tr><td>1</td><td>40°</td><td>?</td></tr><tr><td>2</td><td>110°</td><td>?</td></tr><tr><td>3</td><td>190°</td><td>?</td></tr><tr><td>4</td><td>260°</td><td>?</td></tr><tr><td>5</td><td>330°</td><td>?</td></tr></table>	No.	WCB ($^\circ$)	QB (Quadrantal Bearing)	1	40°	?	2	110°	?	3	190°	?	4	260°	?	5	330°	?	10	CO2	BT5
No.	WCB ($^\circ$)	QB (Quadrantal Bearing)																				
1	40°	?																				
2	110°	?																				
3	190°	?																				
4	260°	?																				
5	330°	?																				

		Part B: Convert Quadrantal Bearings (QB) to Whole Circle Bearings (WCB)																																																	
		<table><tr><th>No.</th><th>QB (Quadrantal Bearing)</th><th>WCB (°)</th></tr><tr><td>6</td><td>N 20° E</td><td>?</td></tr><tr><td>7</td><td>N 50° W</td><td>?</td></tr><tr><td>8</td><td>S 40° W</td><td>?</td></tr><tr><td>9</td><td>S 70° E</td><td>?</td></tr><tr><td>10</td><td>N 15° E</td><td>?</td></tr></table>					No.	QB (Quadrantal Bearing)	WCB (°)	6	N 20° E	?	7	N 50° W	?	8	S 40° W	?	9	S 70° E	?	10	N 15° E	?																											
No.	QB (Quadrantal Bearing)	WCB (°)																																																	
6	N 20° E	?																																																	
7	N 50° W	?																																																	
8	S 40° W	?																																																	
9	S 70° E	?																																																	
10	N 15° E	?																																																	
Q. 4	a	A levelling survey was conducted using the Height of Instrument (HI) method. The Reduced Level (RL) of the Benchmark (BM) is 100 m. Complete the missing values of HI and RL. Give check <table><tr><th>Station</th><th>BS (m)</th><th>IS (m)</th><th>FS (m)</th><th>HI (m)</th><th>RL (m)</th><th>Remarks</th></tr><tr><td>BM</td><td>2</td><td></td><td></td><td>?</td><td>100</td><td>Benchmark</td></tr><tr><td>Point 1</td><td></td><td>3</td><td></td><td></td><td>?</td><td></td></tr><tr><td>Point 2 (CP)</td><td></td><td></td><td>4</td><td></td><td>?</td><td>Change Point</td></tr><tr><td>Point 3</td><td>3</td><td></td><td></td><td>?</td><td>?</td><td>New Instrument Setup</td></tr><tr><td>Point 4</td><td></td><td></td><td>2</td><td></td><td>?</td><td></td></tr></table>					Station	BS (m)	IS (m)	FS (m)	HI (m)	RL (m)	Remarks	BM	2			?	100	Benchmark	Point 1		3			?		Point 2 (CP)			4		?	Change Point	Point 3	3			?	?	New Instrument Setup	Point 4			2		?		10	CO2	BT4
Station	BS (m)	IS (m)	FS (m)	HI (m)	RL (m)	Remarks																																													
BM	2			?	100	Benchmark																																													
Point 1		3			?																																														
Point 2 (CP)			4		?	Change Point																																													
Point 3	3			?	?	New Instrument Setup																																													
Point 4			2		?																																														
Q. 5	a	A surveyor measured a line of 600 m using a chain that was 20 m long. After the work, it was found that the chain was 5 cm too long throughout the measurement. Find the true length of the line.					10	CO1	BT4																																										
Q. 6	a	Explain types of curves with diagram.					10	CO4	BT1																																										
Q. 7	a	Explain remote sensing data acquisition process and platforms and sensors used in remote sensing					10	CO5	BT2																																										
Q. 8	a	Explain Global Positioning Systems with following points 1) Segments 2) GPS measurements 3) errors and biases 4) Surveying with GPS Co-ordinate transformation					10	CO5	BT1																																										
Q. 9	a	Explain electromagnetic spectrum					10	CO5	BT2																																										
Q. 10	a	Explain the working principle of following modern instruments i) Electronic Distance Measurement ii) Total station					10	CO3	BT2																																										